

Report
Of
Cyber Security Internship at Codec Technologies Pvt. Ltd.

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: CS-23411156

Name of Student: AYUSH KUMAR

Batch: 2023-27

CANDIDATE'S DECLARATION

I, Ayush Kumar, do hereby declare that the internship report titled “Cyber Security Internship at Codec Technologies Pvt. Ltd.” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name: AYUSH KUMAR
Roll No.: CS-23411156

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I extend my sincere thanks to Codec Technologies Pvt. Ltd. and Dr. Anurag Shrivastava, Program Manager, for giving me the opportunity to work as a Cyber Security Intern and for their continuous guidance. I am also thankful to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their support.

I express my gratitude to my parents for their blessings.

Signature
Student Name: AYUSH KUMAR
Roll No.: CS-23411156

TABLE OF CONTENTS

Candidate's Declaration	i
Acknowledgement	ii
Internship Completion Certificate.....	—
4. Project Description.....	1
4.1 Introduction.....	1
4.2 Organization Profile	1
4.3 Problem Statement	1
4.4 Project Objectives	1
4.5 Scope of the Project	2
4.6 Technologies and Tools Used	2
4.7 System Architecture.....	2
4.8 Methodology	3
4.9 Expected Outcomes.....	3
4.10 Certificates and Communication Proof.....	4
5. Bibliography / References.....	5

LIST OF TABLES

Table 4.1 Technologies and Tools Used

Table 4.2 Internship Plan (2 Months)

Table 4.3 Expected Outcomes and Skills Gained

4. Project Description

4.1 Introduction

Cyber security is the practice of protecting systems, networks and data from digital attacks. With organisations moving critical services online, the demand for professionals who can find and fix weaknesses before attackers exploit them has grown rapidly. This report describes my two-month internship as a Cyber Security Intern at Codec Technologies Pvt. Ltd., conducted from 25/05/2026 to 25/07/2026 under an AICTE and ICAC approved internship program.

The internship gave me practical exposure to core security activities such as reconnaissance, network scanning, vulnerability assessment, web application testing in a controlled lab, and security reporting, which complemented the theory studied in my B.Tech CSE (Cybersecurity specialization) at IILM University.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is a technology and training company based in Powai, Mumbai, Maharashtra. It is an ISO 9001:2015 certified organization and a Google for Education Partner, and it runs AICTE and ICAC approved internship programs for students in domains such as cyber security. The organization is registered on the National Career Service (NCS) portal (NCS ID: E19E86-011658828892). My internship was supervised by Dr. Anurag Shrivastava, Program Manager.

Website: www.codectechnologies.in | Email: support@codectechnologies.in

4.3 Problem Statement

Most security incidents arise from known weaknesses such as misconfigured services, unpatched software, weak credentials and insecure web applications. Students often learn these concepts only in theory and lack hands-on experience with the tools and processes used by security teams. The problem addressed during this internship was to learn and apply a structured approach to identify, analyse and report security vulnerabilities in a safe, authorised lab environment.

4.4 Project Objectives

- To understand the fundamentals of cyber security, networking and common attack types.

- To gain hands-on experience with industry-standard security tools in an isolated lab.
- To perform reconnaissance, scanning and vulnerability assessment on lab targets.
- To understand common web application risks such as those in the OWASP Top 10.
- To document findings in a clear report with severity ratings and recommended fixes.
- To follow ethical and legal guidelines for security testing.

4.5 Scope of the Project

The work was limited to authorised, intentionally vulnerable lab machines and practice platforms. It covered network and web application security assessment and basic security reporting. Testing on real systems without permission was strictly out of scope, and all activities followed ethical hacking principles.

4.6 Technologies and Tools Used

Table 4.1: Technologies and Tools Used

Category	Tools / Technologies
Operating systems	Kali Linux, Ubuntu, Windows (virtual machines)
Network scanning	Nmap, Wireshark
Vulnerability assessment	OpenVAS / Nessus, OWASP ZAP
Web testing	Burp Suite, browser developer tools
Exploitation (lab only)	Metasploit Framework
Scripting	Python, Bash
Virtualization	VirtualBox / VMware

4.7 System Architecture

The internship work was carried out in an isolated virtual lab consisting of an attacker machine (Kali Linux) and vulnerable target machines connected through a host-only network. This isolation ensured that no real systems were affected. The workflow followed a five-stage security assessment pipeline, shown in Figure 4.1.



Authorised lab environment (isolated virtual machines)

Figure 4.1: Methodology / Workflow of the Internship Project

4.8 Methodology

The internship followed a phased approach, moving from fundamentals to practical assessment and reporting.

Table 4.2: Internship Plan (2 Months)

Period	Activities
Weeks 1-2	Introduction to cyber security, networking basics, Linux, lab setup with virtual machines
Weeks 3-4	Information gathering, port scanning and service enumeration using Nmap; traffic analysis with Wireshark
Weeks 5-6	Vulnerability scanning and analysis; web application testing with Burp Suite and OWASP ZAP
Weeks 7-8	Controlled exploitation in the lab, documentation of findings, remediation suggestions and final report

In each phase, findings were recorded with evidence, rated by severity (Low, Medium, High, Critical) and paired with a recommended mitigation, following the five stages below:

- Reconnaissance: collecting information about the target in the lab network.
- Scanning and enumeration: identifying live hosts, open ports and running services.
- Vulnerability analysis: matching services against known weaknesses and misconfigurations.
- Controlled exploitation: validating selected vulnerabilities only on authorised lab targets.
- Reporting and remediation: documenting results and suggesting fixes such as patching, hardening and input validation.

4.9 Expected Outcomes

Table 4.3: Expected Outcomes and Skills Gained

Outcome	Description
Technical skills	Hands-on use of scanning, analysis and testing tools
Security awareness	Understanding of common vulnerabilities and attack techniques
Reporting skills	Writing structured, evidence-based security reports
Professional ethics	Awareness of legal and ethical boundaries in security testing
Career readiness	Industry exposure and a recommendation letter from the organization

Overall, the internship improved my problem-solving ability, strengthened my understanding of defensive and offensive security concepts, and prepared me for further study and entry-level roles in cyber security.

4.10 Certificates of Completion and Other Communication Proof

The Internship Completion Certificate is attached before Chapter 4. The Letter of Recommendation issued by Codec Technologies Pvt. Ltd. (dated 6/1/2026, MM/DD/YY) is attached below.



5. Bibliography / References

- [1] OWASP Foundation, “OWASP Top 10 Web Application Security Risks,” <https://owasp.org/www-project-top-ten/>.
- [2] Nmap Project, “Nmap Reference Guide,” <https://nmap.org/book/man.html>.
- [3] Wireshark Foundation, “Wireshark User’s Guide,” <https://www.wireshark.org/docs/>.
- [4] PortSwigger, “Web Security Academy,” <https://portswigger.net/web-security>.
- [5] Offensive Security, “Kali Linux Documentation,” <https://www.kali.org/docs/>.
- [6] Codec Technologies Pvt. Ltd., <https://www.codectechnologies.in>.